Leave Management

From a validated customer problem to a shipped, measured feature. This one-pager-per-step walks the whole chain: vision → strategy → business case → build-vs-buy → roadmap → PRD → user stories → cross-team delivery → outcome.

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Product · OVARC

Theme T6
Initiative · Clean payroll closes goal

Release R2 → R3
Feb – Apr 2027 (example)

User research → Vision → Strategy → Business model → Build vs buy → Roadmap → PRD → Epics & stories → Team delivery → Outcome

All names, dates, figures and tools are illustrative examples built on the OVARC case study. The ★ story, OV-801 Request leave, is followed end to end.

VISION & NORTH STAR

VISION

Every company in Egypt pays its people right, on time, with no spreadsheets.

For Leave: leave is requested, approved and paid correctly, with no email threads and no Excel balance sheet to dispute.

NORTH STAR (COMPANY-WIDE)

Clean payroll closes per month

Companies that close monthly payroll on OVARC on time with zero manual corrections afterwards.

WHAT LEAVE MUST NOT BREAK

Every approved or unpaid leave day must reach payroll correctly, or the close isn't clean. Leave is tested against the same bar as payroll itself.

LEAVE'S OWN KEY RESULTS

100%	**90%**	**0**
of leave balances match Law 14 of 2025 entitlements	of leave requests made in the portal, not by phone call to HR	payroll corrections caused by a leave error

> *"HR re-checks every net in Excel before approving and keeps raises in a spreadsheet; errors and forgotten changes are the main fear."*
> U1 · HR and payroll lead interviews, 8 interviews, strength 3 (repeated in support tickets)

STRATEGY CASCADE (PLAYING TO WIN)

Every choice here traces to evidence (U = user research, M = market research) and down to canvas items. Leave is one of four initiatives that serve the company strategy; these are the levels of the cascade that touch it.

WHERE TO PLAY	**W1** SMEs with 20–500 staff in Greater Cairo and Alexandria first	CS1
HOW TO WIN	**H2** Arabic-first and phone-first for employees — this is Leave and the portal's job to prove	VP4 CR2 CS3
HOW TO WIN	**H1** Correctness under Egyptian law is the product, proven by golden tests — Leave inherits this bar from payroll	VP1 KR1
CAPABILITIES	**C1** Payroll rules engine and legal expertise, extended to leave entitlements and accruals	KR1 KR2
MANAGEMENT	**MS** North star and supporting metrics, monthly roadmap review, release exit criteria	VP1

> *"Most staff sign in from phones; HR opens the payroll preview several times per run."*
> U6 · Pilot analytics, 3 pilot clients, strength 2 — behind H2's phone-first bet

WHERE IT SITS IN THE BUSINESS MODEL

The canvas items Leave delivers or depends on, colour-coded by confidence. A full 9-block canvas is in the Business view; this is Leave's slice of it.

VP4
Employees serve themselves in Arabic, on their phone

VP1
Payroll right the first time under Egyptian law

CR2
Self-service portal and SMS for employees

CS1
Egyptian SMEs with 20–500 staff running payroll in Excel

CS3
Employees of client companies (users, not payers)

RS1
Per employee, per month subscription — Leave rides on the same price, doesn't add a line item

Validated · Partly backed · Assumption

WHY LEAVE, AND WHY NOW

Leave is where the clean-payroll promise (VP1) meets the self-service promise (VP4): an unapproved or miscounted leave day is the single most common cause of a payroll correction reported in support tickets (U5). It is sequenced into R2–R3 because payroll's core run (T4) must be correct first.

BUILD VS BUY DECISIONS

Every dependency got one explicit call before design started: buy what's a commodity, build what's the product. These decisions sit in the technical decision map and the sprint dependency list.

Leave and payroll rules engine — BUILD

This is the product. Law 14 entitlements, accrual rules and the balance ledger are OVARC's core IP (H1); no vendor sells "Egyptian labour law as a service".

Capability C1 · KR1 · evidence M1

SMS delivery for approvals and payslip links — BUY

Commodity infrastructure. We use a primary SMS provider with a backup provider and a circuit breaker, rather than operating our own gateway.

Canvas KP2 · decision card L2, dev view

Identity and sign-in (OIDC) — BUY

"Buy an OIDC identity provider; don't store passwords ourselves." Authentication is not where OVARC differentiates, and getting it wrong is a trust risk, not a feature.

Decision card x1, dev view · capability C3

Bank transfer rails — BUY

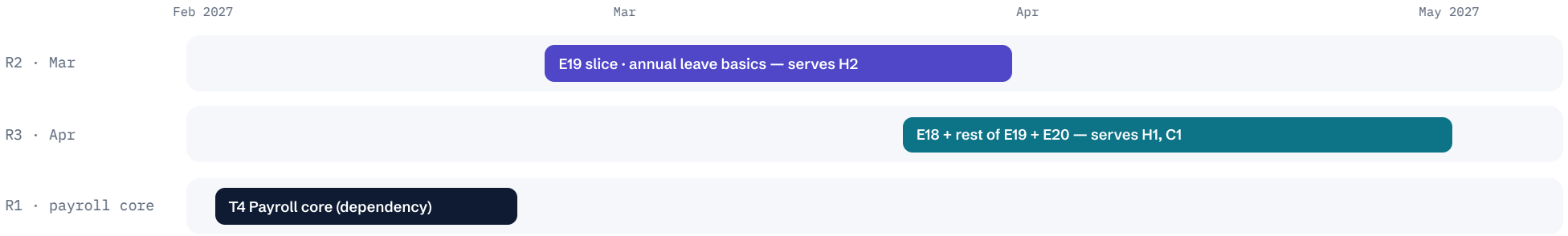
File-based transfer now, bank API once certified. We integrate with existing banking infrastructure rather than building payment rails.

Canvas KP1 · capability C2

> *The rule we apply: if it's regulated, Egyptian and specific to payroll, we build it — that's the moat. If it's generic plumbing (identity, SMS, payments), we buy it and spend the engineering budget on the rules engine instead.*

ROADMAP, CONNECTED BACK TO THE STRATEGY

Leave ships across two releases, not one. Each bar names the strategy choice it serves, not just the epic — so a reader sees why the cut falls where it does.



WHY THIS CUT, NOT "ALL OF LEAVE IN ONE RELEASE"

Payroll needs approved annual leave from February. Accruals and unpaid-leave deductions need payroll rule-engine changes (19 points) and legal sign-off — real work, not padding. Decision record DEC-014 moved that part to R3 rather than slip the whole feature.

THE COST OF THE CUT, MADE EXPLICIT

One month of manual unpaid-leave entry by HR, about 2 hours per client — agreed with the pilot HR lead as acceptable, not assumed by engineering.

PRD — PROBLEM, USERS, SCOPE, NFRS

PROBLEM

Leave lives in email and Excel, so balances are disputed and unpaid leave reaches payroll late or not at all.

USERS

Ahmed, warehouse employee — requests leave, wants to see his balance before sending.
Nour, HR and payroll lead — approves, reconciles the balance ledger against the law.

OBJECTIVE AND KEY RESULTS

Leave is requested, approved and paid correctly without email or Excel. 100% of balances match Law 14; 90% of requests via the portal; zero payroll corrections caused by leave.

In scope

- Law 14 policy defaults per leave type
- Holiday calendar
- Balance ledger and accrual job
- Request, approve, calendar
- Unpaid leave into payroll

Out of scope (this release)

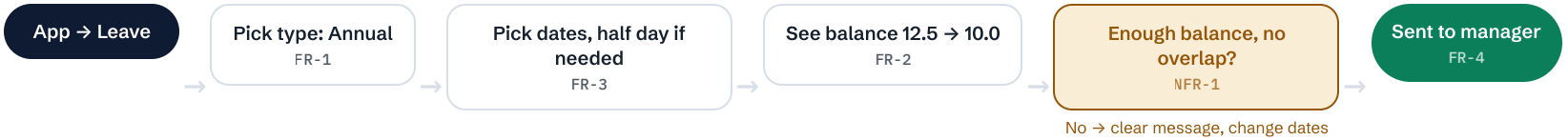
- Shift planning
- Encashment at exit (with offboarding, Later)

NON-FUNCTIONAL REQUIREMENTS

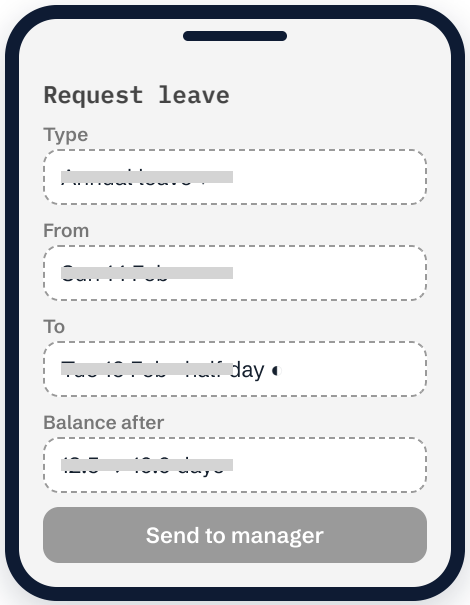
Concurrency	Balance locked while a request is checked; no double spending
Rules	Legal minimums from versioned rules, confirmed by legal counsel
Integration	leave.approved via outbox; payroll consumer idempotent
Privacy	Sick notes restricted; manager sees dates only

The ★ story, followed end to end: the screen Ahmed opens on his phone every time he needs a day off. Below is its user flow, then the same screen from its first grey wireframe to the version tested with 6 warehouse workers.

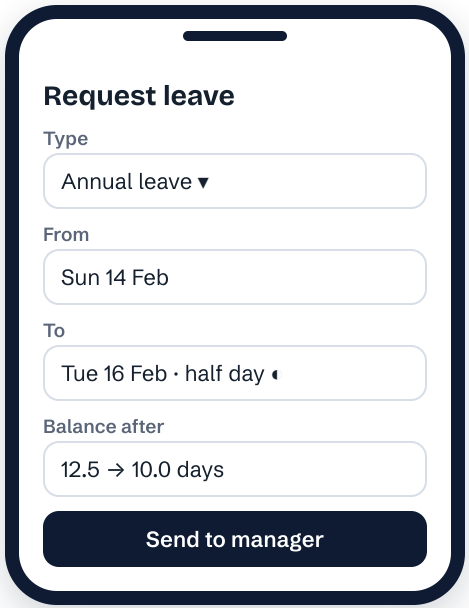
USER FLOW · OV-801 REQUEST LEAVE, FOR AHMED, WAREHOUSE EMPLOYEE



WIREFRAME · LOW FIDELITY



FINAL UI · TESTED WITH USERS



WHAT CHANGED FROM WIREFRAME TO FINAL

- Users asked for full days when they meant half — the half-day toggle moved onto the date itself
- Users wanted the balance before picking dates — balance now shows at the top, then the after-figure
- The button sat below the fold on small phones — it's now a sticky bottom button, reachable by thumb

USABILITY TEST

6 / 6 completed the task, median **41 s**, with 6 warehouse workers, on site, on their own phones. Every fix above came from watching someone's thumb, not a guess.

EPICS → USER STORIES

E18
Leave policies and entitlements
13 PTS · R3

E19
Requests, approvals, calendar
18 PTS · R3

E20
Balances, accruals, payroll impact
19 PTS · R3

THE ★ STORY SPLIT TO FIT THE DEFINITION OF READY (≤ 8 POINTS, TESTABLE)

ID	STORY	PTS	RELEASE
OV-801a	Request annual leave for full days	5	R2
OV-801b	Block requests with no balance or overlap	3	R2
OV-801c	Half-day leave	3	R3
OV-801d	Sick leave with medical note	5	R3
OV-801e	Block dates in a closed payroll month	2	R3

Given a balance of 12.5 days
When Ahmed requests 2.5 days
Then he sees 12.5 → 10.0 before sending

DEFINITION OF READY, CHECKED

Independent · Negotiable · Valuable · Estimable · ≤ 8 points · Testable AC · Design linked · Deps clear

REQUIREMENTS MATRIX: QUALITY BARS × FEATURES × DESIGN

For OV-801 Request leave: every non-functional requirement (rows) is linked to the functional requirements it constrains (dots), to the test or metric that proves it, and to the design element that delivers it. An empty row or column is a gap, fixed before design starts.

ID	QUALITY BAR	TARGET	FR-1 Pick leave type and dates	FR-2 Show balance before and after	FR-3 Allow half days	FR-4 Send to the manager for approval	HOW WE MEASURE IT	WHERE IT SHOWS IN THE DESIGN
NFR-1	**Integrity**	No overlapping leave	●	○	○	○	AC: an overlapping request is rejected and nothing is sent. Pilot: 0 double bookings	Error banner names the clashing date
NFR-2	**Consistency**	Balance can't go negative under parallel requests	○	●	○	○	Concurrency test in CI: two parallel requests, only one succeeds	Balance before → after shown before sending
NFR-3	**Privacy**	Sick-leave notes visible only to HR	●	○	○	○	Role test: a manager sees the dates only, never the note	Note field marked "only HR can see this"
NFR-4	**Mobile**	Works one-handed on a 360 px screen	●	●	●	●	Device test on a 360 px viewport; the button stays visible	Sticky bottom button within thumb reach
NFR-5	**Performance**	Submit answers in under 1 s	○	○	○	●	p95 submit latency in API monitoring	Spinner on Send, success state names the manager

● Bar applies to this feature Measured by QA / monitoring Delivered in the design

HOW THE TEAMS BUILT THIS TOGETHER

One story, five stages, five roles. The UML negotiation produced a written decision record (DEC-014) so the whole team — not just engineering — knew what was deferred and why.

	REFINEMENT	UML NEGOTIATION	DESIGN REVIEW	BUILD & QA	PILOT UAT
Sherif · PM	Walks PRD, splits scope	Negotiates R2/R3 split	—	—	Reads beta signals
Tech lead	Brings sequence diagram	Owns DEC-014	—	Reviews PR	—
Designer	Shows the leave flow draft	—	All 5 states, EN/AR	—	—
Nada · Dev	—	—	—	Builds OV-801, tests	—
Salma · QA	Three amigos, oracles	—	Testability, touch targets	CI gates, exploratory	Sign-off evidence
Nour · HR pilot	—	—	—	—	Runs UAT, signs off

> *DEC-014 · Leave accruals and payroll impact move from R2 to R3 — decided with the tech lead and Nour (HR pilot) in the room, not engineering alone.*

OUTCOME & WHAT I'D TAKE FORWARD

0
double bookings in the pilot

2.4 days
median bug resolution time

100%
P1 test coverage before go-live

≈30→6/wk
leave-related HR calls at the pilot

REFLECTION

The build-vs-buy calls on this feature (buy SMS and identity, build the rules engine) are the ones I'd defend in an interview: they freed the team to spend its scarce engineering time on the one thing a vendor can't sell us — Egyptian labour law, done right. The decision record on deferring accruals to R3 is the other one: it wasn't an engineering call, it was made with HR in the room, because the cost of waiting (one month of manual entry) was theirs to accept, not ours to assume.

— Sherif Hilal, Product